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## SECONDARY BATTERY AND BATTERY MODULE INCLUDING SECONDARY BATTERY


### Abstract

A secondary battery and a battery module including the same are disclosed. In some implementations, the secondary battery may include an electrode assembly including a negative electrode plate, a positive electrode plate, and a separator interposed between the negative electrode plate and the positive electrode plate, a case accommodating the electrode assembly, a vent portion positioned on a plate disposed on one side of the case, the vent portion including a notch portion configured to be breakable; and a leakage prevention portion disposed further outwardly than the vent portion, the leakage prevention portion coupled to the plate on which the vent portion is positioned. The leakage prevention portion may include a leakage prevention layer disposed to oppose the vent portion, a body portion disposed between the leakage prevention layer and the vent portion, the body portion having an opening.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
[0001] This patent document claims the priority and benefits of Korean Patent Application No. 10-2024-0025430 filed on Feb. 22, 2024, the disclosure of which is incorporated herein by reference in its entirety.
TECHNICAL FIELD
[0002] The disclosure and implementations disclosed in this patent document relate to a secondary battery and a battery module including the same.
BACKGROUND
[0003] A secondary battery may be a type of energy storage means that may be charged and discharged. Secondary batteries have been widely used in a variety of electricity-powered devices. For example, secondary batteries have used been as an energy storage means in applications ranging from small devices such as cellphones, laptops, tablets, and the like to large devices such as vehicle, aircraft, and the like. Recently, secondary batteries have been actively researched for use as a vehicle power source.
[0004] Secondary batteries may be categorized into lead-acid batteries, nickel-cadmium batteries, nickel-hydrogen batteries, lithium-ion batteries, and the like, depending on an electrode material. A type of secondary battery may be selected depending on a design capacity, a usage environment, or the like. Alternatively, a secondary battery may be an all-solid-state battery using a solid electrolyte instead of a liquid electrolyte. Li-ion batteries may have relatively high voltages and capacitance, as compared to other types of secondary batteries. As a result, Li-ion batteries have been widely used in devices within fields, requiring high-density energy storage means, such as vehicle battery packs or the like.
[0005] A secondary battery, such as a lithium-ion battery, may include a positive electrode, a negative electrode, a separator, and an electrolyte. The positive electrode and the negative electrode may be disposed with the separator formed of an insulating material interposed therebetween, and may be charged or discharged by the movement of ions through the electrolyte.
[0006] Secondary batteries may be manufactured as flexible pouch-type battery cells or rigid prismatic or cylindrical can-type battery cells.
SUMMARY
[0007] When internal pressure of a secondary battery increases, gas may be discharged through a vent portion formed in a case. During a process of discharging gas, an electrolyte may be discharged together therewith. The discharged electrolyte may affect devices surrounding the secondary battery, and thus it may be necessary to prevent discharge of the electrolyte.
[0008] According to an aspect of the present disclosure, leakage of an electrolyte may be prevented when gas is discharged from a secondary battery.
[0009] According to an aspect of the present disclosure, a secondary battery may have improved stability and safety.
[0010] According to an aspect of the present disclosure, a battery module may have improved stability and safety.
[0011] A secondary battery and a battery module of the present disclosure may be widely applied in the field of green technology, such as to electric vehicles, battery charging stations, and other battery-utilizing solar power generation schemes, wind power generation schemes, or the like. In addition, the secondary battery and the battery module of the present disclosure, may be used in eco-friendly electric vehicles, hybrid vehicles, and the like, to prevent climate change by

suppressing air pollution and greenhouse gas emissions.
[0012] In some embodiments of the present disclosure, a secondary battery may include an electrode assembly including a negative electrode plate, a positive electrode plate, and a separator interposed between the negative electrode plate and the positive electrode plate, a case accommodating the electrode assembly, a vent portion positioned on a plate disposed on one side of the case, the vent portion including a notch portion configured to be breakable; and a leakage prevention portion disposed further outwardly than the vent portion, the leakage prevention portion coupled to the plate on which the vent portion is positioned. The leakage prevention portion may include a leakage prevention layer disposed to face the vent portion, a body portion disposed between the leakage prevention layer and the vent portion, the body portion having an opening.
[0013] In an embodiment, the leakage prevention layer may have a porous structure.
[0014] In an embodiment, the leakage prevention layer may be disposed to be spaced apart from the vent portion by a first distance in a first direction. The first direction may be a direction, perpendicular to one surface of the vent portion.
[0015] In an embodiment, a flat section of the leakage prevention layer may have an area, equal to or greater than an area of a flat section of the vent portion.
[0016] In an embodiment, a flat section of the vent portion may have an oval shape. A flat section of the leakage prevention layer may have an oval shape.
[0017] In an embodiment, the first distance may be equal to or greater than half a second distance. The second distance may be a length of a short axis of the flat section of the vent portion.
[0018] In an embodiment, the leakage prevention layer may be configured to allow gas, discharged from the vent portion, to pass therethrough, and to not allow liquid, discharged from the vent portion, to pass therethrough.
[0019] In an embodiment, the body portion may have a tube shape.
[0020] In an embodiment, the body portion may have a shape gradually widening from the leakage prevention layer to the vent portion.
[0021] In an embodiment, the leakage prevention portion may include a fixing portion protruding along a perimeter of the body portion.
[0022] In an embodiment, the fixing portion may be coupled to and fixed to the plate on which the vent portion is positioned.
[0023] In an embodiment, a plurality of openings may be disposed along a perimeter of the body portion.
[0024] In an embodiment, the openings may be disposed in a distributed manner along the perimeter of the body portion.
[0025] In an embodiment, the body portion may have a first region in which the openings are positioned along the perimeter, and a second region in which the openings are not positioned.
[0026] In another embodiment, the first region may be disposed to face one of side surfaces of the case.
[0027] In another embodiment, the leakage prevention layer may be formed as a plate configured to prevent gas and liquid from passing therethrough. The opening may have a porous structure.
[0028] In some embodiments of the present disclosure, a battery module may include a plurality of secondary batteries, a housing accommodating the plurality of secondary batteries, and a busbar assembly electrically connecting the plurality of secondary batteries to each other. The plurality of secondary batteries may include an electrode assembly including a negative electrode plate, a positive electrode plate, and a separator interposed between the negative electrode plate and the positive electrode plate, a case accommodating the electrode assembly, a vent portion positioned on a plate disposed on one side of the case, the vent portion including a notch portion configured to be breakable, and a leakage prevention portion disposed further outwardly than the vent portion, the leakage prevention portion coupled to the plate on which the vent portion is positioned. The leakage prevention portion may include a leakage prevention layer disposed to face the vent

portion, and a body portion disposed between the leakage prevention layer and the vent portion, the body portion having an opening.
[0029] According to an embodiment of the present disclosure, leakage of electrolyte may be prevented when gas is discharged from a secondary battery.
[0030] According to an embodiment of the present disclosure, a secondary battery may have improved stability and safety.
[0031] According to an embodiment of the present disclosure, a battery module may have improved stability and safety.
[0032] According to an embodiment of the present disclosure, a secondary battery and a battery module may be widely applied in the field of green technology, such as to electric vehicles, battery charging stations, and other battery-utilizing solar power generation schemes, wind power generation schemes, or the like. In addition, according to an embodiment of the present disclosure, the secondary battery and the battery module may be used in eco-friendly electric vehicles, hybrid vehicles, and the like, to prevent climate change by suppressing air pollution and greenhouse gas emissions.

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## Description

BRIEF DESCRIPTION OF DRAWINGS
[0033] Certain aspects, features, and advantages of the present disclosure are illustrated by the following detailed description with reference to the accompanying drawings.
[0034] FIG. **1** is a perspective view of a secondary battery according to the present disclosure.
[0035] FIG. **2** is a cross-sectional view taken along line I-I′ of FIG. **1**.
[0036] FIG. **3** is an enlarged cross-sectional view of portion A of FIG. **2**.
[0037] FIG. **4** is an enlarged cross-sectional view of a state in which gas is discharged from a secondary battery.
[0038] FIG. **5** is a cross-sectional view of a leakage prevention portion.
[0039] FIG. **6** is a plan view of a portion of an top portion of a secondary battery according to the present disclosure.
[0040] FIG. **7** is a plan view of a vent portion.
[0041] FIG. **8** is a plan view of a modification of a body portion.
[0042] FIG. **9** is a perspective view of another modification of a body portion.
[0043] FIG. **10** is a perspective view of another modification of a leakage prevention portion.
[0044] FIG. **11** is a cross-sectional view of a modification of a secondary battery according to the present disclosure.
[0045] FIG. **12** is a perspective view of a battery module according to the present disclosure.
DETAILED DESCRIPTION
[0046] Hereinafter, embodiments of the present disclosure will be described with reference to the attached drawings. For convenience, in the following description, detailed descriptions will be omitted for configurations that obscure the technical gist of the present disclosure or for known configurations.
[0047] The following embodiments are provided to more completely describe the present disclosure to those skilled in the art. The following embodiments are provided to aid understanding of the present disclosure, and the technical idea of the present disclosure is not necessarily limited to particular embodiments described below. The present disclosure should be understood to broadly include all changes, equivalents, and replacements within the idea and the technical scope of the disclosure.
[0048] The terms used herein are provided to more completely describe specific embodiments from the above-described point of view. Accordingly, the terms used herein should not be construed to

reduce, limit, or restrict the technical idea of the present disclosure.

[0049] As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components or a combination thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0050] A secondary battery or a battery cell, described herein, may include a battery that may be charged and discharged. For example, the secondary battery may include a lead acid battery, a nickel cadmium battery, a nickel hydride battery, a lithium-ion battery, and the like. In this description, it is mainly assumed that the secondary battery is a lithium-ion battery. However, it should be understood that a technical concept described herein are applicable to other suitable types of batteries other than a lithium-ion battery.

[0051] The words and terminologies used in the specification and claims should not be construed with common or dictionary meanings, but construed as meanings and conception coinciding the spirit of the present disclosure under a principle that the inventor(s) may appropriately define the conception of the terminologies to explain the present disclosure in the optimum method. Therefore, embodiments described in the specification and the configurations illustrated in the drawings are not more than the most preferred embodiments of the present disclosure and do not fully cover the spirit of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that may replace those when the present application is filed.

[0052] Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In this case, it should be noted that the same components are denoted by the same reference numerals in the accompanying drawings. In addition, detailed descriptions of well-known functions and configurations that may obscure the gist of the present disclosure will be omitted. In addition, some components are exaggerated, omitted, or schematically illustrated in the accompanying drawings, and the size of each component does not fully reflect the actual size. In addition, as used herein, terms such as "top side," "top portion," "top surface," "bottom side," "bottom portion," "bottom surface," and "side surface" are based on the drawings, may vary depending on a direction in which an element or component is actually arranged.

[0053] Hereinafter, a secondary battery and a battery module including the same according to the present disclosure will be described in detail with reference to the drawings.

[0054] FIG. **1** is a perspective view of a secondary battery **100** according to the present disclosure. FIG. **2** is a cross-sectional view taken along line I-I′ of FIG. **1**. FIG. **3** is an enlarged cross-sectional view of portion A of FIG. **2**. FIG. **4** is an enlarged cross-sectional view of a state in which gas is discharged from the secondary battery **100**. FIG. **5** is a cross-sectional view of a leakage prevention portion **200**. FIG. **6** is a plan view of a portion of an top portion of the secondary battery **100** according to the present disclosure. FIG. **7** is a plan view of a vent portion **130**.

[0055] Referring to FIGS. **1** to **7**, the secondary battery **100** may include an electrode assembly **110**, a case **120**, a vent portion **130**, and a leakage prevention portion **200**.

[0056] The secondary battery **100** may include the electrode assembly **110** including a negative electrode plate **115**, a positive electrode plate **116**, and a separator **117** interposed between the negative electrode plate **115** and the positive electrode plate **116**, the case **120** accommodating the electrode assembly **110**, the vent portion **130** positioned on a plate disposed on one side of the case **120**, the vent portion **130** including a notch portion **131** configured to be breakable and the leakage prevention portion **200** disposed further outwardly than the vent portion **130**, and the leakage prevention portion **200** coupled to the plate on which the vent portion **130** is positioned.

[0057] In addition, the leakage prevention portion **200** may include a leakage prevention layer **210** disposed to face the vent portion **130** and a body portion **220** disposed between the leakage prevention layer **210** and the vent portion **130**, and the body portion **220** may have an opening **230**.

[0058] The electrode assembly **110** may include an electrode plate and a separator **117**. The electrode plate may include a positive electrode plate **116** and a negative electrode plate **115**. The separator **117** may be configured as an insulator interposed between the negative electrode plate **115** and the positive electrode plate **116**. The electrode assembly **110** may include a negative electrode plate **115**, a positive electrode plate **116**, and a separator **117** interposed between the negative electrode plate **115** and the positive electrode plate **116**. For example, the electrode assembly **110** may be configured as an assembly-type electrode assembly in which the negative electrode plate **115**, the positive electrode plate **116**, and the separator **117**, interposed between the negative electrode plate **115** and the positive electrode plate **116**, are alternately stacked or arranged. Alternatively, the electrode assembly **110** may be configured as a jelly roll-type electrode assembly in which the negative electrode plate **115**, the positive electrode plate **116**, and the separator **117**, interposed between the negative electrode plate **115** and the positive electrode plate **116**, are alternately stacked or arranged and wound in the form of a roll.

[0059] The case **120** may accommodate the electrode assembly **110**. For example, the case **120** may have an accommodation space **123** therein. The accommodation space **123** may be a space accommodating the electrode assembly **110**.

[0060] The case **120** may include a can **121** and a cap plate **122**. The can **121** may have an opening formed in one side thereof. The can **121** may have a rectangular parallelepiped shape. However, the shape of the case **120** is only an example, and is not necessarily limited to the rectangular parallelepiped shape. The cap plate **122** may seal the opening formed in the one side of the can **121**. For example, the cap plate **122** may be coupled to cover the opening of the can **121**. The cap plate **122** may be coupled to the can **121** to seal the opening. Accordingly, after the electrode assembly **110** is accommodated in the accommodation space **123** formed in the can **121**, the cap plate **122** may be coupled to the can **121** to seal the accommodation space **123**.

[0061] The electrode assembly **110** may be electrically connected to an electrode terminal. The electrode terminal may include a first electrode terminal **113** and a second electrode terminal **114**. The first electrode terminal **113** and the second electrode terminal **114** may be electrically connected to the electrode assembly **110**. For example, the first electrode terminal **113** and the second electrode terminal **114** may be electrically connected to the positive electrode plate **116** and the negative electrode plate **115**, respectively. The positive electrode plate **116** and the negative electrode plate **115** may be electrically connected to the first electrode terminal **113** and the second electrode terminal **114** through a first lead tab **111** and a second lead tab **112**, respectively. At least a portion of the first electrode terminal **113** and the second electrode terminal **114** may be exposed to the outside of the case **120**. For example, the first electrode terminal **113** and the second electrode terminal **114** may pass through the cap plate **122**. Accordingly, at least a portion of the first electrode terminal **113** and the second electrode terminal **114** may protrude toward an top portion of the cap plate **122**.

[0062] The first lead tab **111** may electrically connect the positive electrode plate **116** of the electrode assembly **110** and the first electrode terminal **113** to each other. The first lead tab **111** may include a metal material having electrical conductivity. The first lead tab **111** may be welded or bonded to each of the electrode assembly **110** and the first electrode terminal **113**, such that electricity may flow through the first lead tab **111**.

[0063] One end of the first lead tab **111** may be electrically connected to the electrode assembly **110**, and the other end of the first lead tab **111** may be electrically connected to the first electrode terminal **113**. For example, the first lead tab **111** may have a bent long bar shape. A bottom portion of the first lead tab **111** may be electrically connected to the positive electrode plate **116**, and an top portion of the first lead tab **111** may be connected to the first electrode terminal **113**. However, the shape of the first lead tab **111** is only an example, and may be appropriately changed within the scope of achieving the purpose of the present disclosure.

[0064] The second lead tab **112** may be a component electrically connect the electrode assembly **110** and the second electrode terminal **114** to each other. A shape of the second lead tab **112** may be similar to that of the first lead tab **111**. One end of the second lead tab **112** may be electrically connected to the electrode assembly **110**, and the other end of the second lead tab **112** may be electrically connected to the second electrode terminal **114**.

[0065] The cap plate **122** may have an electrolyte injection hole **124**. The electrolyte injection hole **124** may be a hole passing through the cap plate **122**. An electrolyte may be injected into the can **121** through the electrolyte injection hole **124**. The electrolyte injection hole **124** may be sealed using a stopper or the like after the electrolyte is injected.

[0066] The vent portion **130**, configured to discharge gas into the case **120**, may be positioned on the case **120**. For example, the vent portion **130** may be positioned on a plate disposed on one side of the case **120**. The plate on which the vent portion **130** is positioned may be one of several plates of the case **120**. For example, the plate on which the vent portion **130** is positioned may be the cap plate **122**. Alternatively, the plate on which the vent portion **130** is positioned may be a plate positioned on a side surface or a bottom portion of the can **121**. As described above, the plate on which the vent portion **130** is positioned may be a plate oriented in any direction of the case **120**, and is not necessarily limited to the position illustrated in the drawings.

[0067] The vent portion **130** may include a notch portion **131** configured to be breakable. The notch portion **131** may be a thin portion formed in the vent portion **130**. For example, the vent portion **130** may be configured as a thin plate, and may be coupled to the plate on which the vent portion **130** of the case **120** is positioned, using welding or the like. In this case, the plate on which the vent portion **130** is positioned may have a hole formed in advance, such that the vent portion **130** may be coupled thereto. The notch portion **131** may be a portion of the vent portion **130**, formed of a thinner material or another material that is prone to breakage. The notch portion **131** may have weak rigidity than other portions of the vent portion **130**. That is, at a pressure higher than or equal to a pressure set for the vent portion **130** to break, the notch portion **131** may break before the other portions of the vent portion **130**. A shape of the notch portion **131** may vary, and is not necessarily limited to the shape of the notch **131** illustrated in the drawings.

[0068] The leakage prevention portion **200** may be configured to discharge gas discharged from the vent portion **130**, and to prevent liquid from being discharged. The leakage prevention portion **200** may be disposed further outwardly than the vent portion **130**. “Outwardly” referred to herein may be a direction toward the outside of the case **120**. Conversely, the accommodation space **123** may be positioned in the case **120**. The leakage prevention portion **200** may be disposed at a position relatively farther from the accommodation space **123** than the vent portion **130**. For example, the accommodation space **123**, the vent portion **130**, and the leakage prevention portion **200** may be disposed from the inside to the outside of the case **120**.

[0069] The leakage prevention portion **200** may include a leakage prevention layer **210** and a body portion **220**.

[0070] The leakage prevention layer **210** may be disposed to face the vent portion **130**. For example, the leakage prevention layer **210** may be configured as a thin film or a plate. The leakage prevention layer **210** may be configured as a thin layer, and may have a surface. At least one surface of the leakage prevention layer **210** may face at least one surface of the vent portion **130**. The leakage prevention layer **210** may be positioned further outwardly than the vent portion **130** in a state of facing the vent portion **130**.

[0071] The leakage prevention layer **210** may have a porous structure. For example, the leakage prevention layer **210** may have a mesh structure. The leakage prevention layer **210** may ensure air permeability through the porous structure.

[0072] The leakage prevention layer **210** may be configured to allow gas discharged from the vent portion **130** to pass therethrough, and to not allow liquid to pass therethrough. When gas is discharged from the vent portion **130**, an electrolyte, accommodated in the case **120**, may also be discharged from the vent portion **130**. The electrolyte may include a material having high ionic conductivity or a highly acidic material. Accordingly, when the electrolyte is completely discharged to the outside of the secondary battery **100**, a chemical reaction may occur with other secondary batteries **100** or devices positioned on the outside of the secondary battery **100**. A hole having a porous structure may have a large size sufficient to externally discharge gas. However, the size may be small to discharge liquid. For example, the leakage prevention layer **210** may include a membrane. The membrane may be an ultra-thin film manufactured by stretching expanded Polytetrafluoroethylene (e-PTFE) under specific conditions. A hole of the membrane, having a porous structure, may have a large size sufficient to externally discharge gas. However, the size may be small to discharge liquid.

[0073] The leakage prevention layer **210** may be disposed to be spaced apart from the vent portion **130** by a first distance d**1** in a first direction Z. The first direction Z may be a direction, perpendicular to one surface of the vent portion **130**. Specifically, the leakage prevention layer **210** may be disposed on the outside of the case **120** than the vent portion **130** in the first direction Z. Accordingly, gas or liquid, discharged from the vent portion **130**, may be in a certain space positioned between the leakage prevention layer **210** and the vent portion **130**. The first direction Z may change depending on a position of the vent portion **130**. For example, when the vent portion **130** is positioned on a side surface of the can **121** in an X-axis direction or a Y-axis direction, the first direction Z may also be the X-axis direction or Y-axis direction. That is, the first direction Z is not necessarily limited to the Z-axis direction illustrated in the drawings. The first distance d**1** may be a distance between the leakage prevention layer **210** and the vent portion **130**.

[0074] An area of a flat section of the leakage prevention layer **210** may be equal to or greater than an area of a flat section of the vent portion **130**. The flat section may be a section of the leakage prevention layer **210** or the vent portion **130** in top view. For example, in top view, the vent portion **130** may be positioned on the inside of a region of the leakage prevention layer **210**. However, the present disclosure is not limited thereto, and the flat section of the leakage prevention layer **210** may be formed to be smaller than the flat section of the vent portion **130** within the scope of achieving the purpose of the present disclosure.

[0075] The flat section of the vent portion **130** may have an oval shape, and the flat section of the leakage prevention layer **210** may have an oval shape. The oval shape referred to herein may also include an oval shape having a straight line. For example, the flat section of the vent portion **130** may have top and bottom portions having a straight line, and left and right sides having a round semicircular shape. The flat section of the leakage prevention layer **210** may have a shape similar to that of the vent portion **130**. For example, the flat section of the leakage prevention layer **210** may be an enlarged shape of the flat section of the vent portion **130**.

[0076] The first distance d**1** may be appropriately adjusted depending on a size of the vent portion **130**, a shape of the notch portion **131**, or the like. For example, the first distance d**1** may be equal to or greater than half a second distance d**2**. The second distance d**2** may be a length of a short axis L**2** of the flat section of the vent portion **130**. When the flat section of the vent portion **130** has an oval shape, the flat section may have a short axis L**2** and a long axis L**1**. The long axis L**1** may be a long virtual line connecting central points of left and right edges, among edges of the vent portion **130**, to each other, and the short axis L**2** may be a virtual line connecting central points of top and bottom edges, among the edges of the vent portion **130**, to each other. The flat section of the vent portion **130** may be symmetrical about the long axis LI or the short axis L**2**. In this case, a length of the short axis L**2** may be defined as the second distance d**2**.

[0077] At least a portion of the vent portion **130** may be upwardly tilted when the notch portion **131** is broken by gas pressure. In this case, it may be necessary to maintain an appropriate distance such that a portion of the vent portion **130** does not hit the leakage prevention layer **210**, positioned thereabove. For example, when the notch portion **131** is formed along the long axis L**1** of the vent portion **130**, a height at which the vent portion **130** is tilted may be half of the second distance d**2**. Accordingly, the first distance d**1** may be at least half of the second distance d**2**. However, the first distance d**1** may be adjusted depending on a shape of the notch portion **131** or a degree to which the vent portion **130** is tilted, and is not limited thereto.

[0078] The body portion **220** may be disposed between the leakage prevention layer **210** and the vent portion **130**. For example, the leakage prevention layer **210** and the vent portion **130** may be disposed to be spaced apart from each other in a state of opposing each other. The body portion **220** may be disposed in a space between the leakage prevention layer **210** and the vent portion **130**. The body portion **220** may be coupled to the leakage prevention layer **210** and formed integrally with the leakage prevention layer **210**. For example, the body portion **220** may have a form in which a circumference or perimeter of the leakage prevention layer **210** extends in a direction of the vent portion **130**.

[0079] Specifically, the body portion **220** may have a tube shape. The tube shape may have an empty space through which gas or liquid flows, and may be a shape including a wall surrounding the empty space. The body portion **220** may have an internal empty space to allow gas or liquid to pass therein. For example, gas or liquid, discharged from the vent portion **130**, may move to the internal empty space of the body portion **220**.

[0080] The body portion **220** may have a shape gradually widening from the leakage prevention layer **210** to the vent portion **130**. For example, the flat section of the body portion **220** may have a narrow top portion and a wide bottom portion. Referring to FIG. **3**, a side section of the body portion **220** may have a shape similar to a trapezoidal shape. The flat cross section may be a section of the body portion **220** in top view, and the side section may be a section of the body portion **220** in side view. The body portion **220** may have a wide bottom portion, and may be stably coupled to the case **120**. In addition, a volume of the leakage prevention portion **200**, protruding toward the outside of the case **120**, may be reduced, thereby efficiency of a space of a secondary battery. However, the shape of the body portion **220** described above is only an example. Accordingly, the top portion and the bottom portion of the body portion **220** may be formed to have the same sectional area. Conversely, the top portion may be formed to have a sectional area, greater than that of the bottom portion.

[0081] The leakage prevention portion **200** may be coupled to a plate on which the vent portion **130** is positioned. The leakage prevention portion **200** may receive force due to gas discharged from the vent portion **130**. Accordingly, the leakage prevention portion **200** may be fixed to and coupled to the plate on which the vent portion **130** is positioned, and thus may not be separated from the case **120**.

[0082] The leakage prevention portion **200** may include a fixing portion **240**. The fixing portion **240** may protrude along a perimeter of the body portion **220**. For example, the fixing portion **240** may be in the form of one end of the body **220** protruding outwardly from the body **220**. The fixing portion **240** may protrude in all directions of the perimeter of the body portion **220**. The fixing portion **240** and the body portion **220** may form a certain angle. The leakage prevention portion **200** may have a shape similar to a shape of a hat. Referring to FIG. **5**, a direction in which the fixing portion **240** protrude may be the same as a direction of a plane of the leakage prevention layer **210**. The direction in which the fixing portion **240** protrudes and the angle, formed by the fixing portion **240** and the body portion **220**, are not limited, and the fixing portion **240** may be appropriately changed within the scope in which the fixing portion **240** is inserted into and coupled to the plate on which the vent portion **130** is positioned.

[0083] The fixing portion **240** may be coupled to the plate on which the vent portion **130** is positioned to be fixed to the plate on which the vent portion **130** is positioned. The fixing portion **240** may have a sufficiently large area to be coupled to the plate on which the vent portion **130** is positioned. The fixing portion **240** may be inserted into the plate on which the vent portion **130** is positioned, and may be coupled to the plate on which the vent portion **130** is positioned using a method such as welding. A groove having a shape, corresponding to a shape of the fixing portion **240**, may be formed in advance in the plate on which the vent portion **130** is positioned, such that the fixing portion **240** may be inserted into the groove. The fixing portion **240** may be inserted into the groove and then coupled to the groove using a method such as welding.

[0084] The body portion **220** may have an opening **230**. The opening **230** may be a hole formed in the body **220**. Gas, discharged from the vent portion **130**, may also be discharged through the opening **230**. A shape, a quantity, and a direction of the openings **230** may be formed in various manners.

[0085] For example, a plurality of openings **230** may be disposed along the perimeter of the body portion **220**. Specifically, the openings **230** may be disposed in a distributed manner along the perimeter of the body portion **220**. The plurality of openings **230** may be disposed at regular intervals. In this case, gas, discharged from the vent portion **130**, may be discharged through the opening **230**, and may be evenly spread. A pressure or an amount of the discharged gas may be divided and reduced through several openings **230**.

[0086] Gas, discharged from the vent portion **130**, may be in contact with the leakage prevention layer **210** corresponding to a primary discharge direction. At least a portion of gas may be discharged through the leakage prevention layer **210**, and may be secondarily discharged in a distributed manner through the opening **230**. The dotted arrow illustrated in FIG. **4** indicates a path through which gas is discharged. In this case, an electrolyte that may be discharged together with gas through the vent portion **130** may not pass through the leakage prevention layer **210**, and may be blocked. Accordingly, the leakage prevention layer **210** may prevent the electrolyte from being discharged out of the secondary battery **100**. In this case, a small amount of electrolyte may be discharged through the opening **230**.

[0087] FIG. **8** is a plan view of a modification of the body portion **220**. FIG. **9** is a perspective view of another modification of the body portion **220**.

[0088] Referring to FIGS. **8** and **9**, an opening **230** may be disposed in a certain region.

[0089] The body portion **220** may have a first region S**1** in which the opening **230** is positioned along a perimeter thereof, and a second region S**2** in which the opening **230** is not positioned. For example, the body portion **220** may be divided into a first region S**1** and a second region S**2** along the perimeter thereof. Other openings **230** may be disposed at regular intervals between a first opening **230** and a last opening **230**. The first region S**1** may be a region having the first opening **230** to the last opening **230**. In this case, the first region S**1** may also include a region positioned between a plurality of openings **230**. However, there may be a relatively widest region between the first and last openings **230**, and the region may not be included in the first region S**1** but may be the second region S**2**.

[0090] The second region S**2** may be a region excluding the first region S**1**. The body portion **220** may have various numbers of first regions S**1** and second regions S**2**. For example, the number of first regions S**1** and the number of second regions S**2** may be one, respectively. FIG. **8** illustrates one first region S**1** and one second region S**2**. FIGS. **1** to **6** illustrate only a first region S**1**. Conversely, there may be a plurality of first regions S**1** and a plurality of second regions S**2**. FIG. **9** illustrates two first regions S**1** and two second regions S**2**.

[0091] The first region S**1** may be disposed to face one of the sides of the case **120**. In this case, the opening **230** may be disposed to face a specific direction. For example, the first region S**1** may be disposed to face one of the four sides of the case **120**. Referring to FIG. **8**, the sides of the case **120** may have four sides opposing the X-axis or Y-axis. When the opening **230** is disposed to face a specific direction, the first region S**1** may specify the direction in which gas is discharged through the opening **230**.

[0092] FIG. **10** is a perspective view of another modification of a leakage prevention portion **200***b*.

[0093] Referring to FIG. **10**, a leakage prevention layer **210***b* may be formed as a plate, and an opening **230***b* may have a porous structure.

[0094] The leakage prevention layer **210***b* may be formed as a plate configured to prevent gas and liquid from passing therethrough. For example, the leakage prevention layer **210***b* may not have a porous structure. In this case, neither gas nor liquid may pass through the leakage prevention layer **210***b*. In this case, gas or an electrolyte, discharged from the vent portion **130**, may first collide with the leakage prevention layer **210***b* and then secondarily move toward the opening **230***b*.
[0095] The opening **230***b* may have a porous structure. The porous structure may be formed to allow gas to pass therethrough but not liquid to pass therethrough. Accordingly, gas, secondarily moving toward the opening **230***b*, may pass through the porous structure, and may be discharged out of the secondary battery **100**. However, an electrolyte may not pass through the porous structure and may collide once again. For example, the opening **230***b* may include the membrane described above.
[0096] FIG. **11** is a cross-sectional view of a modification of the secondary battery **100** according to the present disclosure.
[0097] Referring to FIG. **11**, a leakage prevention portion **200** may be positioned on a bottom portion of a case **120**.
[0098] A plate on which a vent portion **130**, on which the leakage prevention portion **200** is positioned, is positioned may be one of all plates of the case **120**. FIG. **10** illustrates a case in which the plate on which the vent portion **130** is positioned is positioned on the bottom portion of the case **120**. A position of the plate on which the vent portion **130** is positioned is not limited to the position illustrated in the drawings of the present disclosure.
[0099] FIG. **12** is a perspective view of a battery module **10** according to the present disclosure.
[0100] Referring to FIG. **12** together with FIGS. **1** to **11**, the battery module **10** may include a plurality of secondary batteries **100**, a housing **20**, and a busbar assembly **30**. For example, the battery module **10** may include the plurality of secondary batteries **100**, the housing **20** accommodating the plurality of secondary batteries **100** and the busbar assembly **30** electrically connecting the plurality of secondary batteries **100** to each other. And the plurality of secondary batteries **100** may include an electrode assembly **110** including a negative electrode plate **115**, a positive electrode plate **116**, and a separator **117** interposed between the negative electrode plate **115** and the positive electrode plate **116**, a case **120** accommodating the electrode assembly **110**, a vent portion **130** positioned on a plate disposed on one side of the case **120**, and including a notch portion **131** configured to be breakable and a leakage prevention portion **200** disposed further outwardly than the vent portion **130**.
[0101] The leakage prevention portion **200** coupled to the plate on which the vent portion **130** is positioned, and the leakage prevention portion **200** may include a leakage prevention layer **210** disposed to face the vent portion **130** and a body portion **220** disposed between the leakage prevention layer **210** and the vent portion **130**. In addition, the body portion **220** may have an opening **230**.
[0102] A module referred to herein may be based on a concept including a general module and pack. Accordingly, the battery module **10** may also refer to a battery pack.
[0103] The plurality of secondary batteries **100** may be a unit in which a plurality of secondary batteries **100** are aggregated. Each secondary battery **100** may be one of the secondary batteries **100** illustrated in FIGS. **1** to **10**.
[0104] The housing **20** may accommodate the plurality of secondary batteries **100**. To accommodate the plurality of secondary batteries **100**, the housing **20** may include a bottom frame **21**, a partition wall **22**, and a cover **23**. The bottom frame **21** may have a space in which the plurality of secondary batteries **100** are accommodated. Frames, positioned on a bottom portion and side surfaces of the module, may be gathered to form the bottom frame **21**. The partition wall **22** may be a wall for partitioning an internal space of the bottom frame **21** into several zones. The plurality of secondary batteries **100** may be accommodated in respective spaces partitioned by the partition wall **22**. The cover **23** may be coupled to an top portion of the bottom frame **21**. The cover

**23** may cover the plurality of secondary batteries **100** accommodated in the bottom frame **21**.
[0105] The busbar assembly **30** may be configured to electrically connect the plurality of secondary batteries **100** to each other. An arrangement and a shape of the busbar assembly **30** are not limited, and the busbar assembly **30** may be a component electrically connecting the plurality of secondary batteries **100** to each other. The busbar assembly **30** may be coupled to the housing **20**. For example, the busbar assembly **30** may be in a state of being coupled to the cover **23**, opposing electrode terminals of a secondary battery.
[0106] Only specific examples of implementations of certain embodiments are described. Variations, improvements and enhancements of the disclosed embodiments and other embodiments may be made based on the disclosure of this patent document.

## Claims

**1**. A secondary battery comprising: an electrode assembly including a negative electrode plate, a positive electrode plate, and a separator interposed between the negative electrode plate and the positive electrode plate; a case accommodating the electrode assembly; a vent portion positioned on a plate disposed on one side of the case, the vent portion including a notch portion configured to be breakable; and a leakage prevention portion disposed further outwardly than the vent portion, the leakage prevention portion coupled to the plate on which the vent portion is positioned; wherein the leakage prevention portion includes: a leakage prevention layer disposed to face the vent portion; and a body portion disposed between the leakage prevention layer and the vent portion, the body portion having an opening.
**2**. The secondary battery of claim 1, wherein the leakage prevention layer has a porous structure.
**3**. The secondary battery of claim 2, wherein the leakage prevention layer is disposed to be spaced apart from the vent portion by a first distance in a first direction, and the first direction is a direction, perpendicular to one surface of the vent portion.
**4**. The secondary battery of claim 3, wherein a flat section of the leakage prevention layer has an area, equal to or greater than an area of a flat section of the vent portion.
**5**. The secondary battery of claim 3, wherein a flat section of the vent portion has an oval shape, and a flat section of the leakage prevention layer has an oval shape.
**6**. The secondary battery of claim 5, wherein the first distance is equal to or greater than half a second distance, and the second distance is a length of a short axis of the flat section of the vent portion.
**7**. The secondary battery of claim 1, wherein the leakage prevention layer is configured to allow gas, discharged from the vent portion, to pass therethrough, and to not allow liquid, discharged from the vent portion, to pass therethrough.
**8**. The secondary battery of claim 1, wherein the body portion has a tube shape.
**9**. The secondary battery of claim 8, wherein the body portion has a shape gradually widening from the leakage prevention layer to the vent portion.
**10**. The secondary battery of claim 8, wherein the leakage prevention portion includes a fixing portion protruding along a perimeter of the body portion.
**11**. The secondary battery of claim 10, wherein the fixing portion is coupled to and fixed to the plate on which the vent portion is positioned.
**12**. The secondary battery of claim 1, wherein a plurality of openings are disposed along a perimeter of the body portion.
**13**. The secondary battery of claim 12, wherein the openings are disposed in a distributed manner along the perimeter of the body portion.
**14**. The secondary battery of claim 12, wherein the body portion has a first region in which the openings are positioned along the perimeter, and a second region in which the openings are not positioned.

**15**. The secondary battery of claim 14, wherein the first region is disposed to face one of side surfaces of the case.

**16**. The secondary battery of claim 1, wherein the leakage prevention layer is formed as a plate configured to prevent gas and liquid from passing therethrough, and the opening has a porous structure.

**17**. A battery module comprising: a plurality of secondary batteries; a housing accommodating the plurality of secondary batteries; and a busbar assembly electrically connecting the plurality of secondary batteries to each other, wherein the plurality of secondary batteries include: an electrode assembly including a negative electrode plate, a positive electrode plate, and a separator interposed between the negative electrode plate and the positive electrode plate; a case accommodating the electrode assembly; a vent portion positioned on a plate disposed on one side of the case, the vent portion including a notch portion configured to be breakable; and a leakage prevention portion disposed further outwardly than the vent portion, the leakage prevention portion coupled to the plate on which the vent portion is positioned, and the leakage prevention portion includes: a leakage prevention layer disposed to face the vent portion; and a body portion disposed between the leakage prevention layer and the vent portion, the body portion having an opening.